STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	**CASE 23-G-0225**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service.	**CASE 23-G-0226**

TESTIMONY OF MARGOT SPINDELMAN

Dated: September 1, 2023

Q. Please state your name, address, and identify for whom you are presenting testimony in these proceedings.

A. My name is Margot Spindelman, my address is 106 Clay St, Brooklyn, NY 11222 and I am presenting on my own behalf as a rate payer and community member in National Grid's KEDNY territory, who is, along with my neighbors, directly affected by The Company's application to fund the extension of fossil fuel infrastructure.

Q. Have you previously testified before the New York State Public Service Commission?

A. This is my first time testifying. To quote Rory Christian, Chair of the New York Public Service Commission and CEO of the New York Department of Public Service, from his recent article "Harmonizing Utility Regulation's Stakeholders" in the August 2023 edition of *Fortnightly Magazine*, "By embracing the growing desire for engagement, Public Utility Commissions and Boards have the opportunity to shape policies that truly reflect the needs and aspirations of the general public." With my testimony I embrace that objective.

Q. What is the purpose of your testimony in these proceedings?

1. My testimony is meant to highlight National Grid's continued focus on extending our gas infrastructure while simultaneously undercutting efforts that would offer carbon- and methane-free alternatives.

2. New York State's climate goals as codified in the Climate Leadership and

Community Protection Act ("CLCPA") mandate that we reduce emissions and facilitate the transition to the use of clean energy. If this rate case is fully funded it will instead perpetuate our reliance on fossil fuel.

3. National Grid has been mandated to implement at least one Utility Thermal Energy Network ("U-TEN") pilot but has offered an insufficient and incomplete plan, in stark contrast to their robust filing in this rate case that aims to keep gas flowing by patching instead of decommissioning pipelines. Evidence of this is based on the Company's initial pilot filing and a Scoping Study/PEER Review of Proposed U-TEN for Brooklyn NYCHA submitted by Egg Geo.

Q. Are you sponsoring any exhibits?

A. Yes. I am sponsoring (3) exhibits as follows:

- Exhibit_(MS-01): National Grid's filing in Case 23-00902/23-G-0225 requesting approval for five FTE positions to implement the U-TEN pilot (page 85, The Brooklyn Union Gas Company d/b/a National Grid NY, Direct Testimony of the Gas Infrastructure and Operations Panel, April 28, 2023).

- Exhibit_(MS-02): Case 22-M-0429 Scoping Study / PEER Review of Proposed U-TEN for Brooklyn NYCHA submitted by Egg Geo, Case 22-M-0429, filed September 1, 2023

- Exhibit_(MS-03): National Grid Utility Thermal Energy Network Pilot Proposal, Case 22-M-0429, filed January 9, 2023

Q. What is the relevance of National Grid Utility Thermal Energy Network ("U-TEN") Pilot Proposal, Case 22-M-0429 to this rate case, Case 23-00902/23-G-0225?

A. While cost recovery of the U-TEN pilot in Case 22-M-0429 is not part of this proceeding, full time employees for that pilot are being requested in this rate case. The Company sets out its request for funding for five FTEs to design, develop, install, inspect, repair and manage its U-TEN Pilot (Case 22-M-0429) located in Brooklyn at a New York City Housing Authority ("NYCHA") complex at Vandalia Avenue.

Testimony of the Gas Infrastructure and Operations Panel

On January 9, 2023, the Company submitted a thermal energy network pilot proposal in Case 22-M-0249 ("UTEN Pilot"). The Company is seeking approval for cost recovery of the UTEN Pilot in Case 22-M-0429 and not in this proceeding. In this case, the Company is seeking five FTE positions to support the successful implementation of the UTEN Pilot. Of the five FTEs requested, the Company needs: (1) three FTEs in gas asset engineering to design and develop networked systems, work closely with contractors, and conduct outreach to partners and customers; (2) one FTE for a maintenance mechanic responsible for installation, inspection, and repair of the installed networked system(s); (3) 0.5 FTE for a lead program manager to deliver the program and prepare reports and analytics; and (4) 0.5 FTE for a lead project manager who will have responsibility for the networked thermal energy project from project approval, implementation, development, execution, and closure.

Exhibit_(MS-01)

Q. In what ways was National Grid's Utility Thermal Energy Network Pilot Proposal insufficient and incomplete?

A. Thermal energy networks extract, deliver, balance, and recycle heat among buildings and other sources and sinks— heat that would otherwise be wasted. Heat recovery and recycling makes these systems extremely efficient. Heat is not created in these systems, but simply transferred. Thermal energy networks are able to provide both heating and cooling using a fraction of the energy that is used when we burn fossil fuels. New York State passed the Utility Thermal Energy Network and Jobs Act in June, 2022, and the Governor signed it into law on July 5, to move ahead quickly with this energy saving, greenhouse gas emission slashing, life enhancing technology. The seven largest utilities in the State were tasked with creating and implementing pilots to provide important learnings on how to build these systems at scale.

National Grid's initial submission for a pilot in their KEDNY territory, specifically at the New York City Housing Authority ("NYCHA") complex at Vandalia Avenue, fails in several ways.

1. KEDNY's preferred design is for heating only.

   KEDNY proposes a pilot at a NYCHA complex at 17, 47, and 77 Vandalia Avenue in Brooklyn, and suggests possibly including nearby commercial sites, to date unnamed in their filings. The total connected load is estimated to serve 417,000 square feet. As pointed out in the Egg Geo "Case 22-M-0429 Scoping Study / PEER Reveiw of Proposed U-TEN for Brooklyn NYCHA", "Grid's consultant's analysis reported, as noted above, that the cost of providing heating and cooling to the NYCHA buildings is estimated at over $67.7 million, compared to an estimate of slightly over $38 million if heating only is provided to the NYCHA buildings. The Company states

that this is due to the considerable internal upgrades that would be required to add cooling to these specific buildings, specifically to replace the risers. National Grid originally had intended to propose both heating and cooling for all buildings. Considering the significant costs for upgrading the NYCHA buildings, National Grid believes it will be preferable to only include heating for the NYCHA buildings. National Grid said that it will work with NYCHA to determine what solutions and energy efficiency incentives may be available and preferable to ensure that residents have efficient, cost-effective energy systems to meet their needs. We note that National Grid's upstate projects include both heating and cooling at much lower estimated costs" (Exhibit_ (MS-02), page 3)

The Study goes on to say that "this project does not presently meet the minimum criteria to be considered a thermal energy network ("TEN")." (Exhibit_(MS-02), page 4)

2. By aiming for a heating only scenario, tenants may be responsible for providing cooling, which would be a signifcant burden on the tenants who live in what is classified as a Disadvantaged Community ("DAC").

"It is not clear if the $38 million estimate includes the cost of alternative cooling systems that would be required, such as the window air-source heap pumps (ASHP) that NYCHA and NYSERDA have been developing through the Clean Heat for All Challenge. These ASHPs also provide heat, so their use would duplicate the TEN heating function, albeit less efficiently and with exposure to weather. It is also unclear if residents would bear the cost for operating the air conditioners." (Exhibit_(MS-02), page 5)

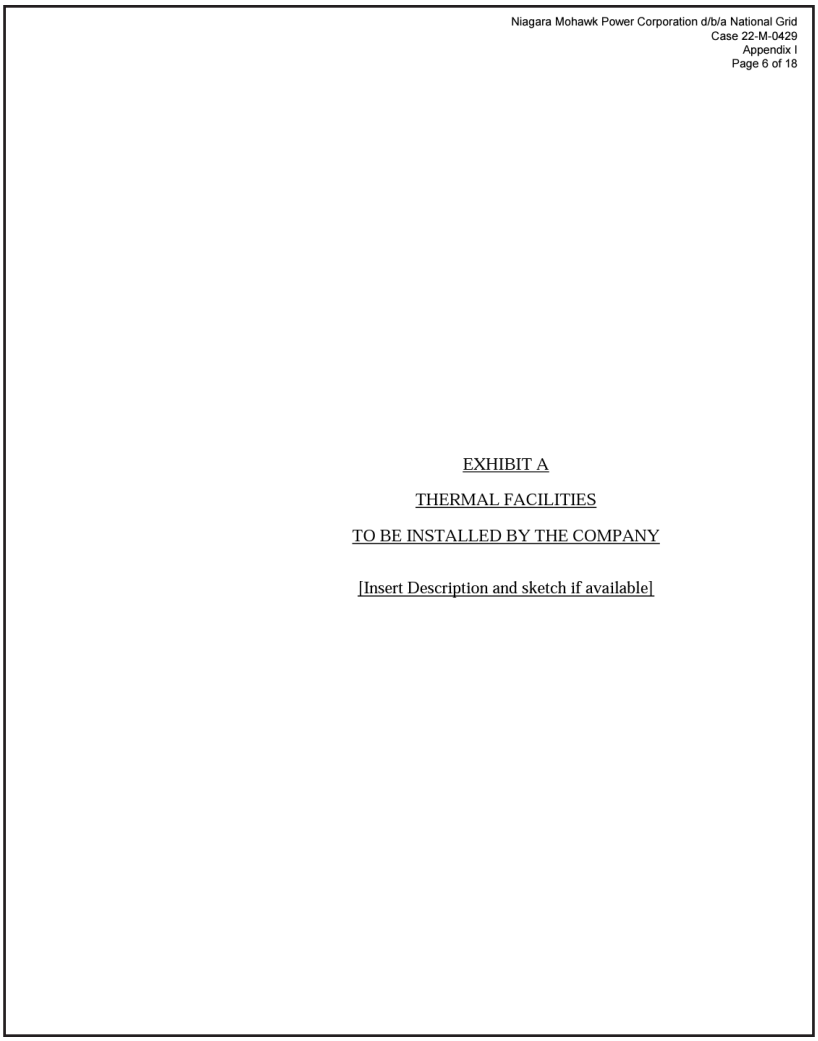
3. The Egg Geo report describes alternative, nearby sources of heat, negating the need for an extensive and expensive geothermal borefield. "The Vandalia Avenue NYCHA facility is located in an area rich with thermal sources and sinks that could be used to develop an exemplary thermal energy pilot. For example, the NYC Transit Authority (NYCTA) of the MTA pumps large quantities of groundwater on a daily basis to dewater around two nearby subway stations. The pumped water is discharged into the tidal canal east of the Vandalia Avenue NYCHA facility." The study continues, "We suggest the Company engage MTA to utilize this potential thermal resource." And, "In addition, a standard surface water plate exchanger configuration, ideally located in the tidal channel west of the facility, would allow a significant reduction in geothermal boreholes needed on the project, eliminating the need for so many boreholes and substantially reducing the cost of the project" "Finally, the project is located adjacent to the city's 26th Ward Wastewater Resources Recovery Facility (WRRF) where thermal exchange using Wastewater Energy Transfer (WET) could easily be implemented" (Exhibit_ (MS-02), page 8)

4. The study goes on to cast doubt on the high cost the Company cites as a reason to omit cooling as part of the pilot. They write "Our engineers and scientists have a reasonable degree of certainty that even if the entire system had to be replaced with a new networked geothermal heating and cooling system in the buildings, the pricing shared by the Company is unusually high, probably due in part to the pricing of a bore-field that likely was designed to handle the entire heating load with little or no diversity." (Exhibit_(MS-02),

page 9)

5. While the Company does allude to diversifying the load (which is required to create a successful TEN) by adding commercial properties, they neglect to specify any partners in this project by name. The Egg Geo report notes "Without a diversified load (meaning a load that has different usage characteristics from the anchor property), the geothermal exchanger will take the entire wintertime heating load without any other source of waste heat from adjacent properties. This project should only be commissioned as a pilot when at least one commercial property with a properly diversified load has been identified and the owners of that property have provided a written commitment to participate in the project. That could be any number of adjacent commercial properties, which would enjoy substantial savings in heating and cooling costs by connecting with the TEN. Such a commitment is required to meet the definition of a Thermal Energy Network (TEN)" (Exhibit_(MS-02), page 7)

6. Pages of the submitted appendices were titled with headings indicating key elements of the proposal, but were blank, showing what can only be called a shocking lack of commitment to the effort. Although these pages come from a sample Customer Agreement and Protection Plan, they fail to detail any specifics by which an actual customer scenario can be assessed.



Exhibit_(MS-03) Appendix I page 6

EXHIBIT B

CUSTOMER EQUIPMENT

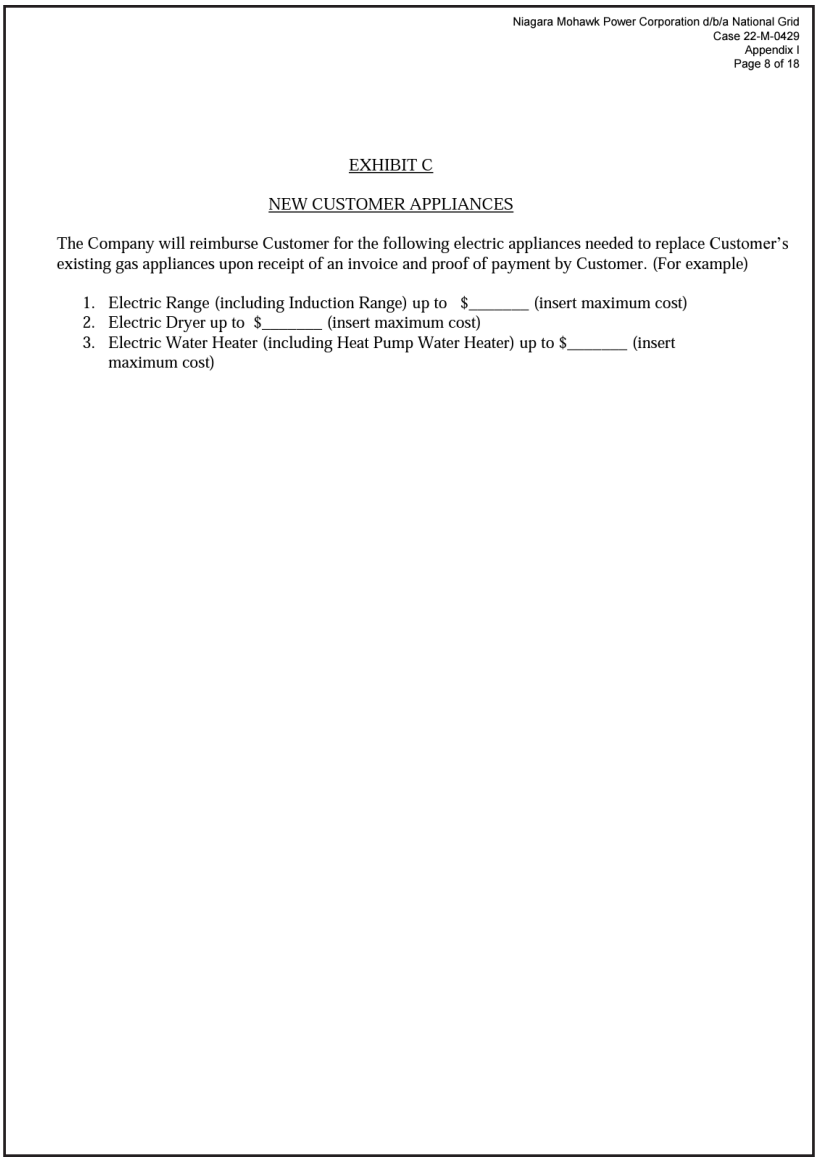
1. Customer Equipment that preexisted the Program

[Describe the Customer equipment and heating distribution system that pre-existed the Program]

2. Customer Equipment to be installed as part of the Program

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EXHIBIT C

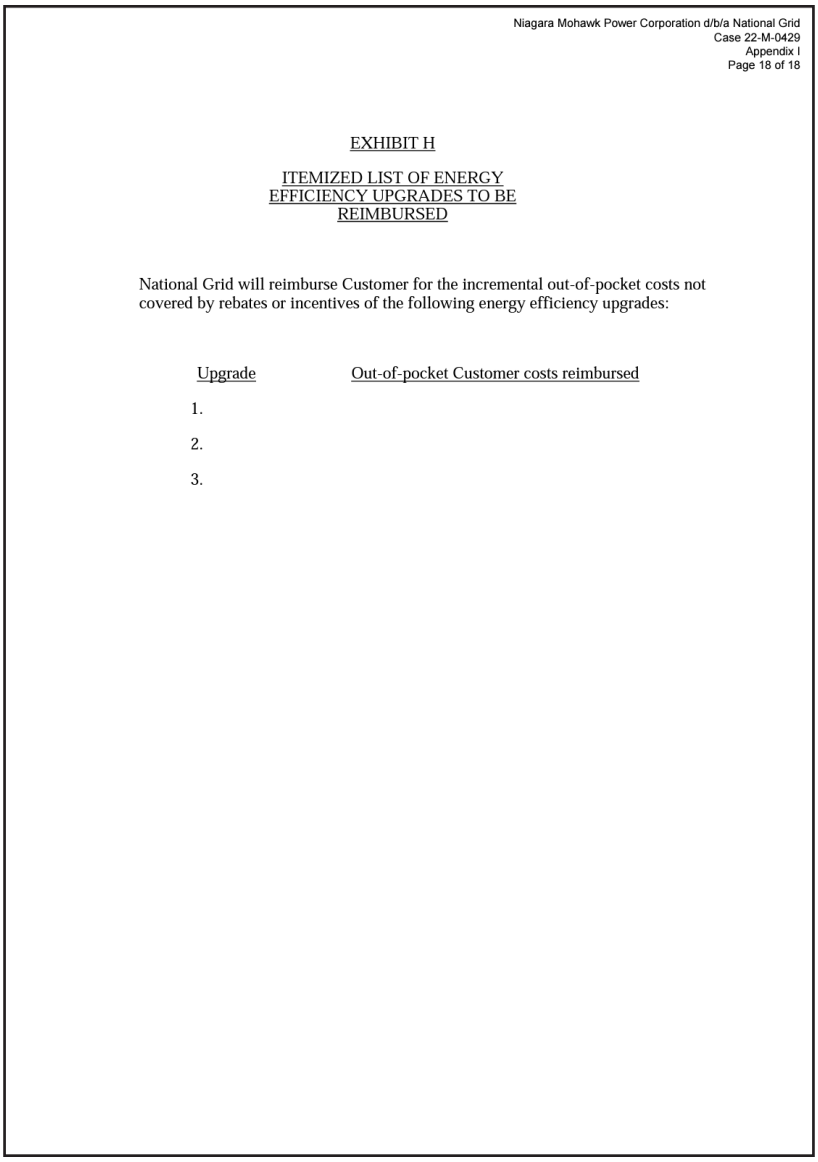
NEW CUSTOMER APPLIANCES

The Company will reimburse Customer for the following electric appliances needed to replace Customer's existing gas appliances upon receipt of an invoice and proof of payment by Customer. (For example)

1. Electric Range (including Induction Range) up to $_____ (insert maximum cost)
2. Electric Dryer up to $_____ (insert maximum cost)
3. Electric Water Heater (including Heat Pump Water Heater) up to $_____ (insert maximum cost)

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EXHIBIT H

ITEMIZED LIST OF ENERGY EFFICIENCY UPGRADES TO BE REIMBURSED

National Grid will reimburse Customer for the incremental out-of-pocket costs not covered by rebates or incentives of the following energy efficiency upgrades:

Upgrade	Out-of-pocket Customer costs reimbursed
1.	
2.	
3.	

Exhibit_(MS-03) Appendix I page 18

7. While National Grid is proposing to exclude cooling from their pilot (at a time when excessive heat is becoming a persistent and dangerous reality that threatens the health and safety of many New Yorkers) Egg Geo suggests that a properly designed TEN that includes both heating and cooling as well as

hot water can be built for roughly the same cost as the Company's heating only plan. "We estimate that the entire project will cost less than $40,000,000 for heating and cooling with central geothermal heat pumps and fan coils in each apartment. Using MTA and surface water exchange will further reduce these costs as it will reduce or eliminate the need for a geothermal bore field. Wastewater Energy Transfer (WET) can further help to reduce thermal loads for Domestic Hot Water (DHW), heating, and cooling by exchanging thermal energy with the wastewater as it leaves the apartment complex." (Exhibit_ (MS-02, page 14)

Q. Recognizing the difference in scale of a proposal for a single U-TEN project vs a rate case filing for the entire downstate territory, how do the requests compare?

A. The meager effort of KEDNY to rise to the challenges and opportunities presented by the state mandated U-TEN proposal is in vivid contrast to their detailed and comprehensive request for massive amounts of money ($3,801,356,821) to expand and extend the life of their gas system in myriad ways. By law and public demand in the face of the climate emergency, it is inevitable that gas systems will be phased out. KEDNY should recognize that its survival depends on shifting its business model from piping methane to piping water for thermal energy networks. The UTENJA law threw gas utilities a lifeline: the Company should seize it and release the gas infrastructure that will pull it under.